

✓ COMPLETE - DEPLOYED TO PRODUCTION

FLeT REST API Implementation Debrief

Deployment Overview & Technical Architecture

DATE: January 19-20, 2026

VERSION: v1.4.52

PACKAGE: mgraph_ai_service_html_graph

STATUS: Live



EXECUTIVE STATUS: LIVE & OPERATIONAL

LOC. [1,1] STAT. [2,2]

STATUS CONSOLE	
ENVIRONMENT:	Dev (html-graph.dev.mgraph.ai) ✓ LIVE
DOCUMENTATION:	Swagger Docs ✓ AVAILABLE

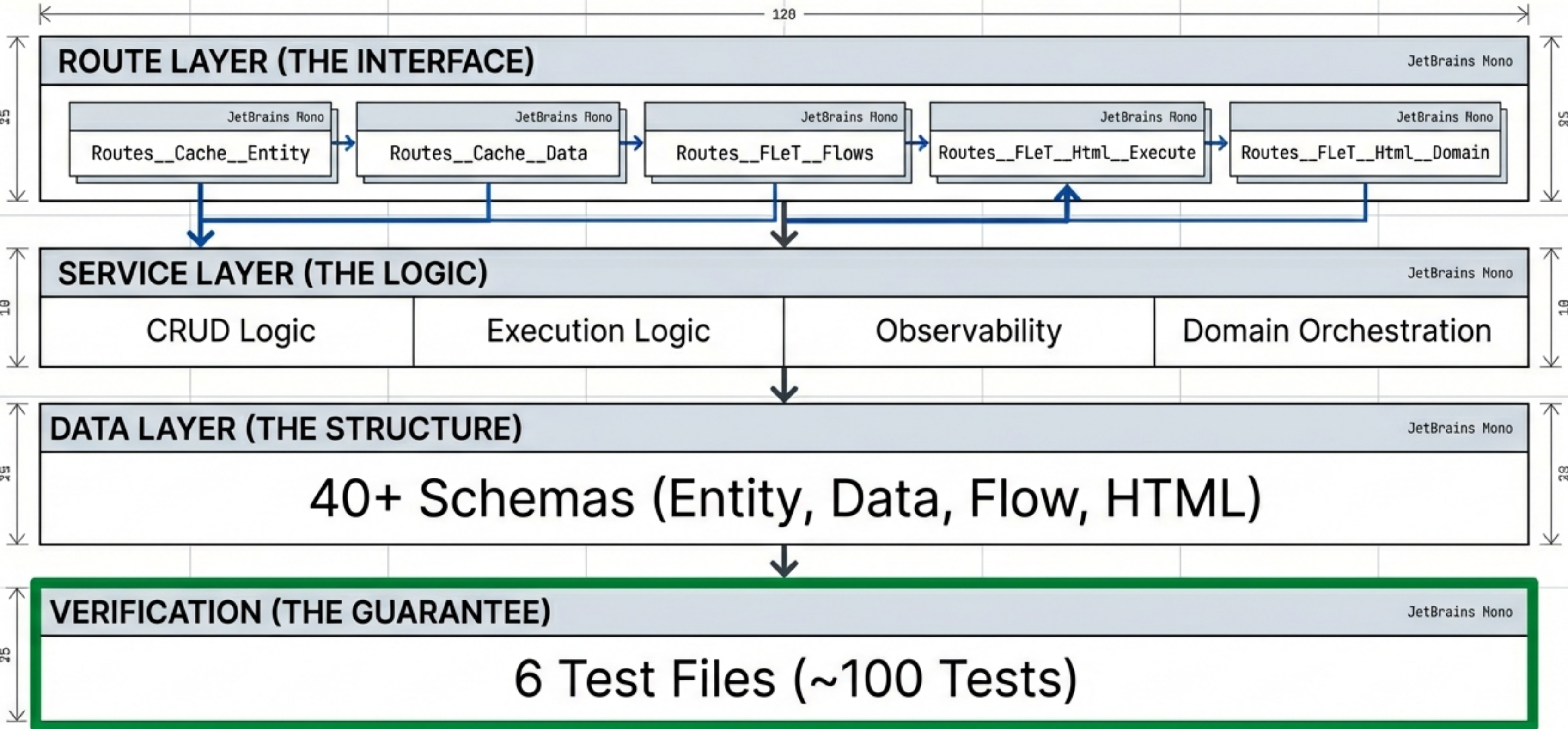
LOC. [1,1] STAT. [2,2]

Primary Deliverable: A fully Type_Safe, production-ready REST API for HTML graph operations.

METRIC 001	METRIC 002	METRIC 003	METRIC 004
JetBrains Mono	JetBrains Mono	JetBrains Mono	JetBrains Mono
5	5	40+	100%
SERVICE CLASSES	ROUTE CLASSES	SCHEMA CLASSES	TEST PASS RATE
Type_Safe implementation	FastAPI endpoints	Request/response DTOs	~100 cases across 6 files

SYSTEM ANATOMY & LAYERED COMPONENTS

LOC. [1,1] STAT. [2,2]



ALIGNMENT WITH REQUIREMENTS

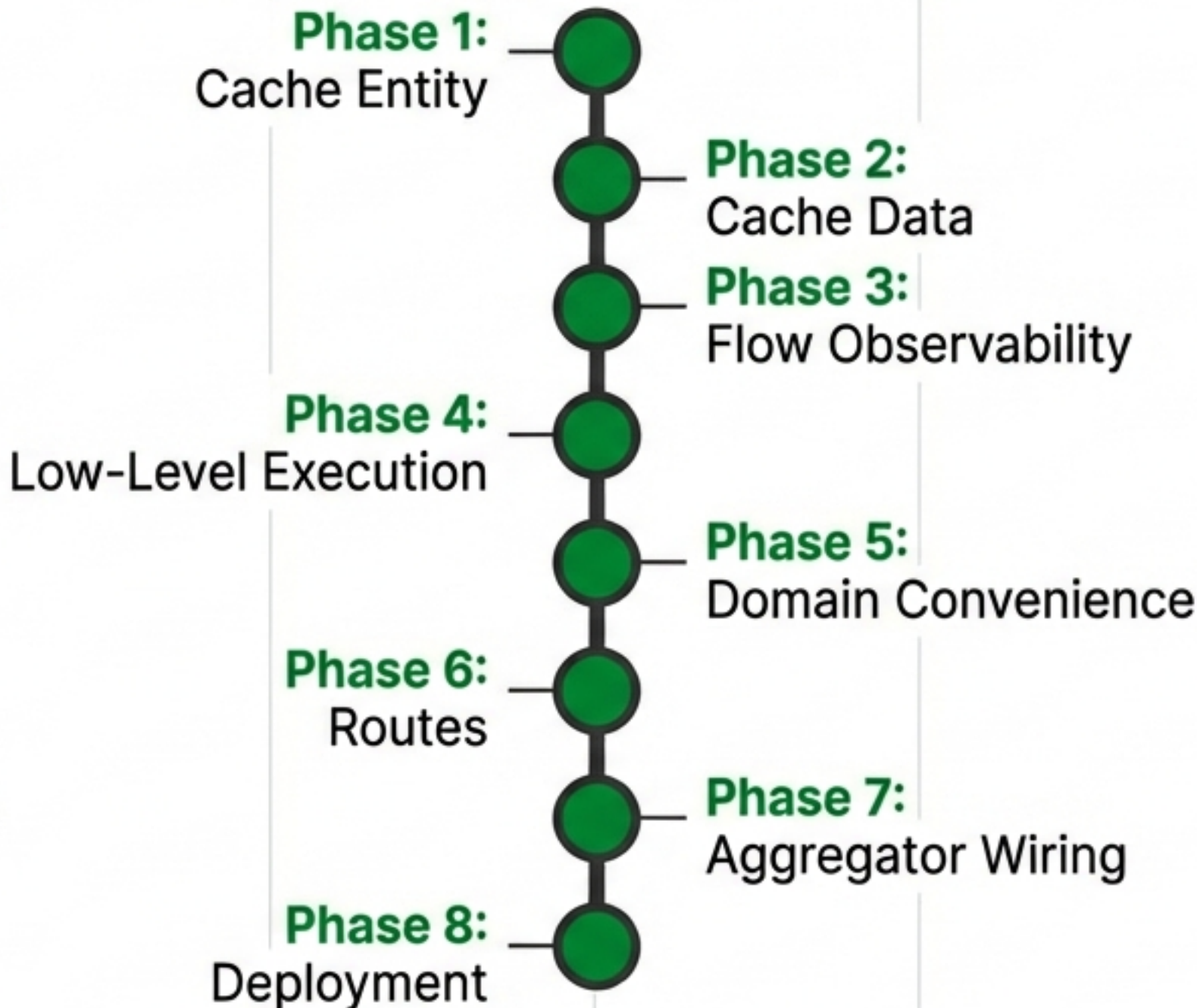
LOC. [1,5]

STAT. 13,21

Brief vs. Implementation

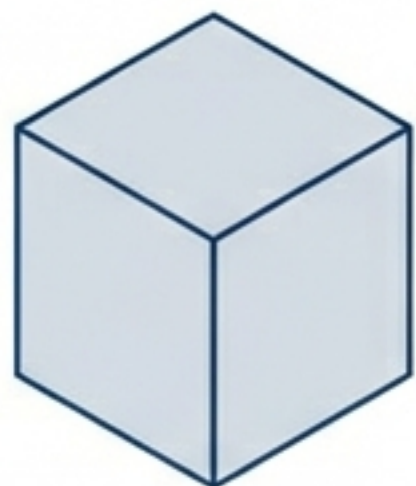
- Route: `/cache/{namespace}/entity/*`
JetBrains Mono ☒ Complete
- Route: `/cache/{namespace}/data/{cache_id}/*`
JetBrains Mono ☒ Complete
- Route: `/flet/flows/*` & `/flet/html/*`
JetBrains Mono ☒ Complete
- Design: Thin routes, rich services
Inter ☒ Implemented
- Standard: Full Type_Safety throughout
Inter ☒ Implemented

Phase Completion Tracking



KEY ARCHITECTURAL DECISIONS

Real Objects in Tests



Real Object



Mock

Rationale: Using a fast in-memory cache enables true integration testing without the fragility of mocks.

Entity Semantics

entry__store

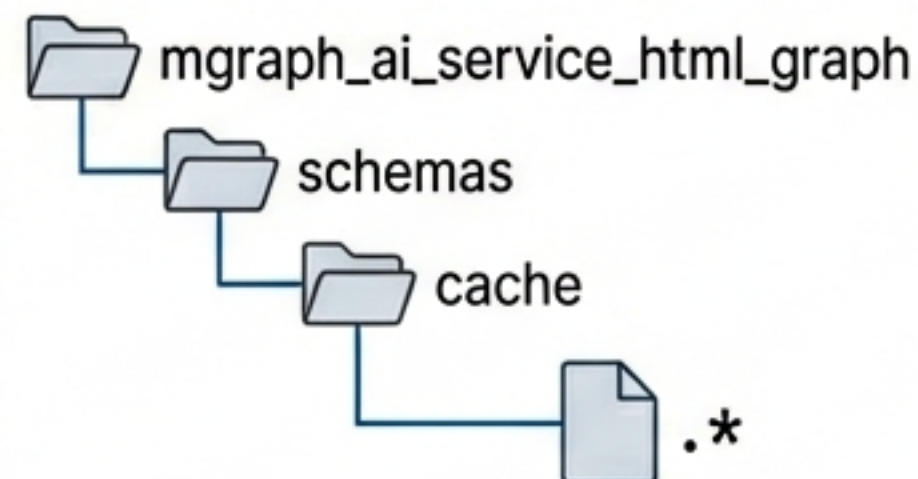
JetBrains Mono



New
Entity

Rationale: "entry__store" always creates new entities. Enables strict versioning via "cache_hash" tracking.

Schema Organization



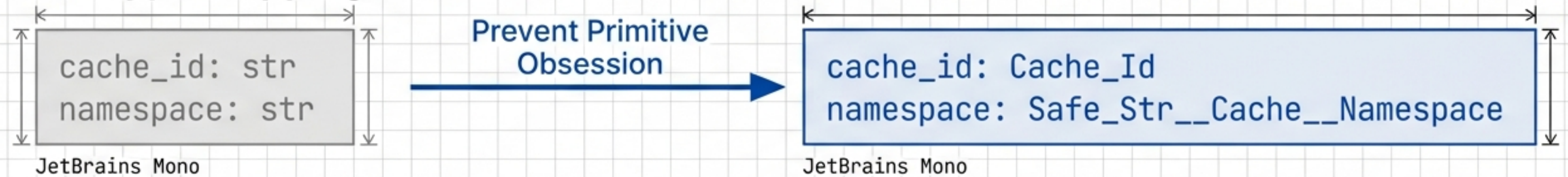
Rationale: Consistent import paths matching existing package structures.

TECHNICAL SPOTLIGHT: SAFETY & STORAGE

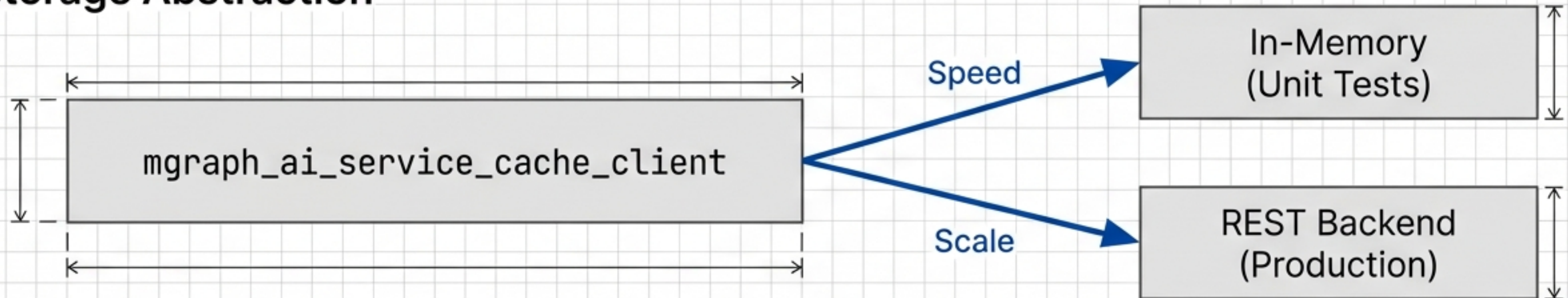
LOC. [1,5]

STAT. [3,2]

Safe Type Mappings



Storage Abstraction



Exact same service code runs in both environments via environment variables.

Solved Challenge: URL to Cache Key

LOC. [2,1]

STAT. [4,7]

The Solution:

Url__To__Cache_Key

Input

https://site.
com?
id=123&track=
xyz&sort=asc

JetBrains Mono

Output

Safe_Str__Cache
__File__Cache_Key

JetBrains Mono

- Configurable Param Handling
- Sanitization & Sorting
- Hashing Modes

JetBrains Mono

Solving the problem of converting inconsistent, messy URLs into hierarchical, predictable storage keys.

Testing Strategy: The 'No Mocks' Philosophy

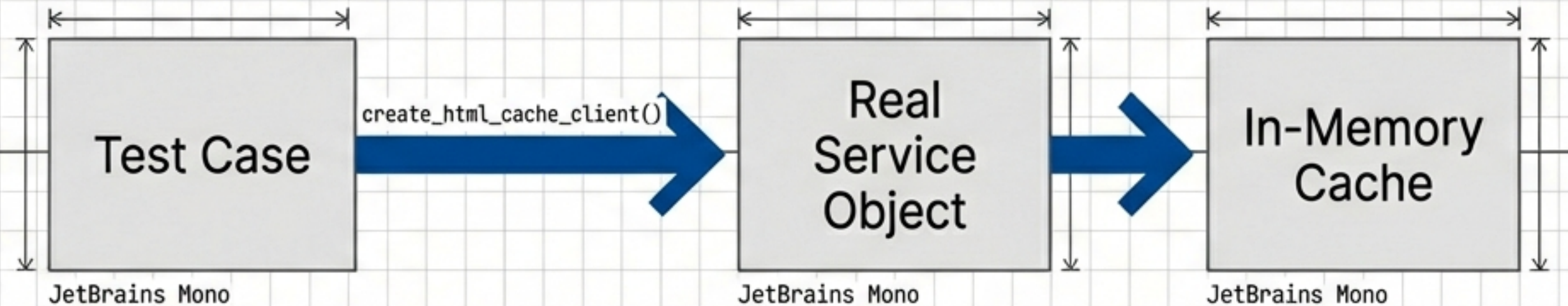
LOC. [2,1]

STAT. [4,7]

~100

Total Tests

Cache__Entity__Service: 17
Cache__Data__Service: 23
FleT__Html__Domain__Service: 21
Url__To__Cache_Key: 30
Others: ~10



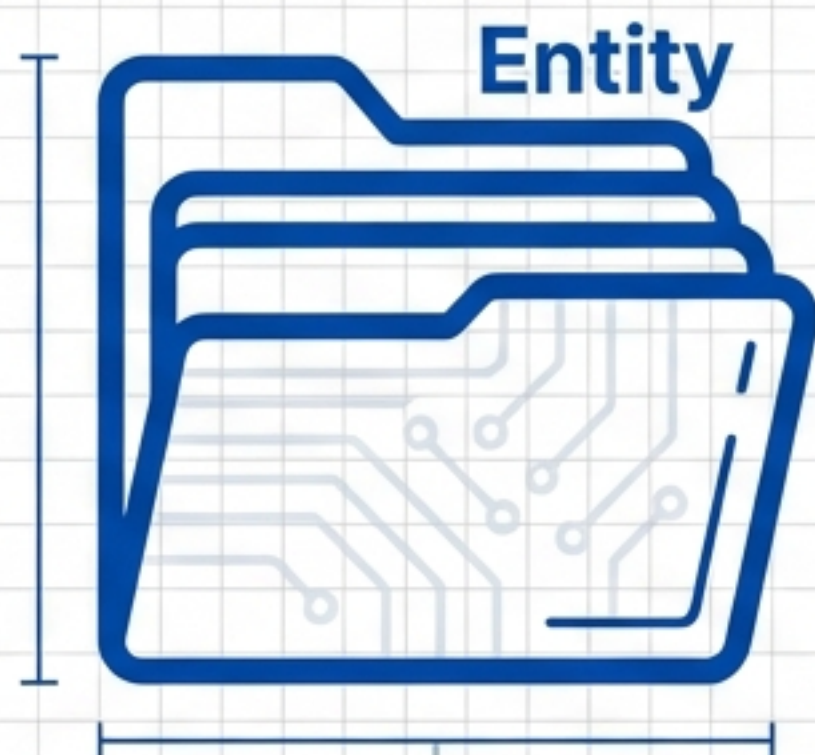
Zero mock
maintenance
overhead.

Catches real
integration issues.

Tests serve as accurate documentation.

API Capability: Cache Entity Management

Base Path: /cache-entity/



Key Endpoints

- **POST** /{namespace}/entity/create
Always creates a new entity (Versioning).
- **POST** /{namespace}/entity/lookup
Find by key or hash.
- **GET** /{namespace}/entity/{cache_id}/metadata
Retrieve metadata without heavy payload.
- **GET** /{namespace}/entity/{cache_id}/refs
View relationships.

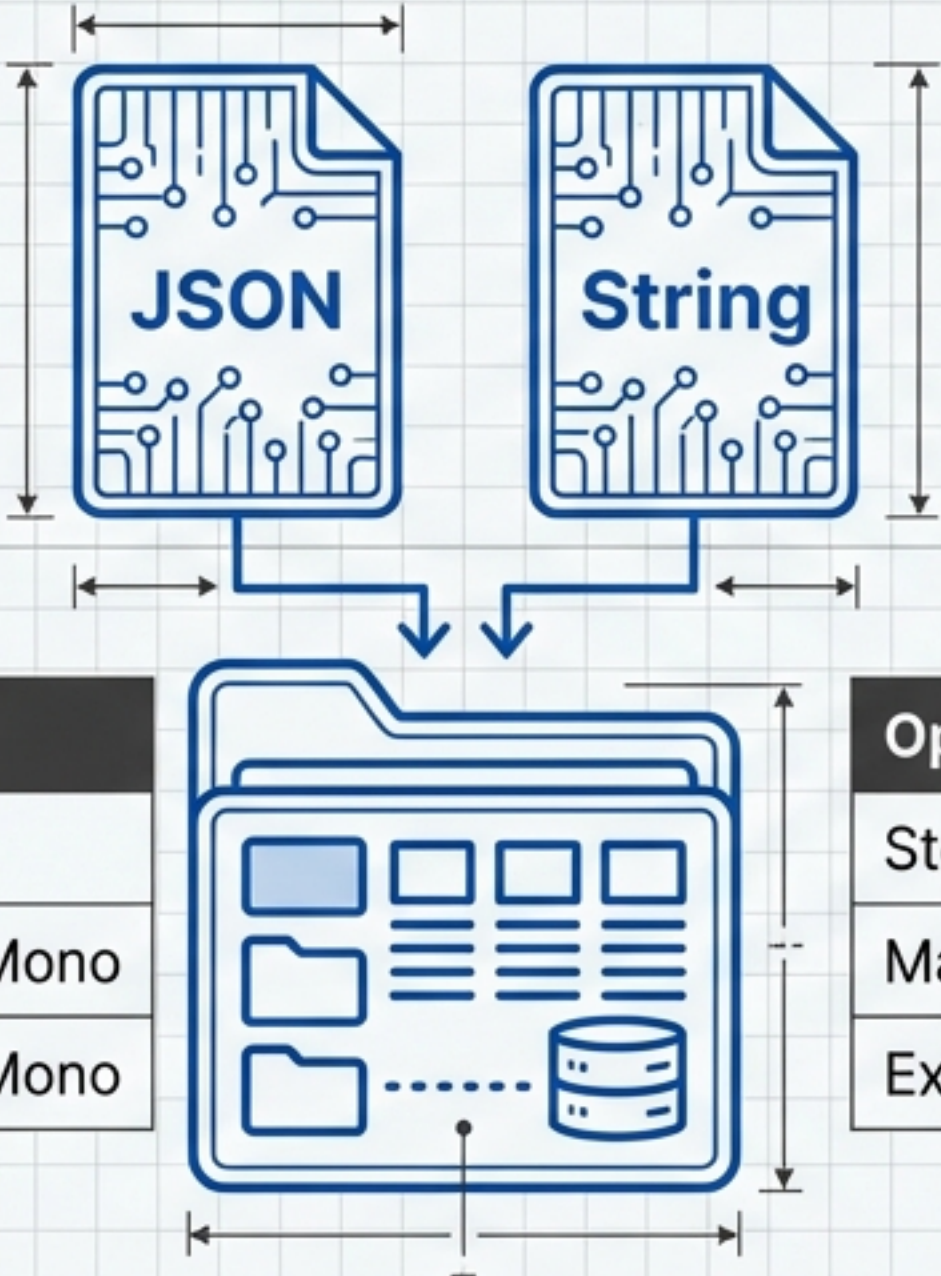
Core Concept: Separation of '**Entity**'
(**Metadata**) from '**Data**' (**Content**).



API Capability: Data & File Storage

STAT. [5,2]

Base Path: /cache-data/



Operation
Inter
JetBrains Mono
JetBrains Mono

Operation	Methods
Store & Retrieve	store_string, store_json, get_string, get_json
Management	list_files, list_paths, delete_all
Existence	exists (checks specific keys)

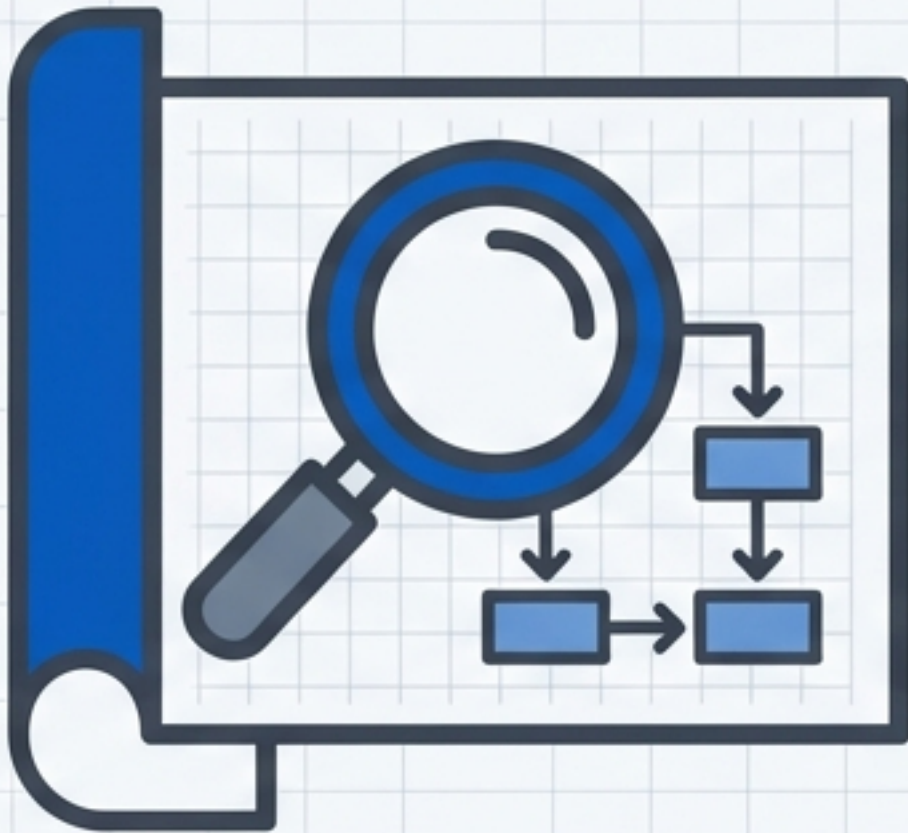
✓ Security: Uses 'Safe_Str__Cache__File__Data_Key' and 'File_Id' to strictly prevent path traversal issues.

LOC. [3,1]

API Capability: FLeT Flows & Observability

Base Path: /flet-flows/

Value Proposition: Full transparency into the 'Black Box' of execution.



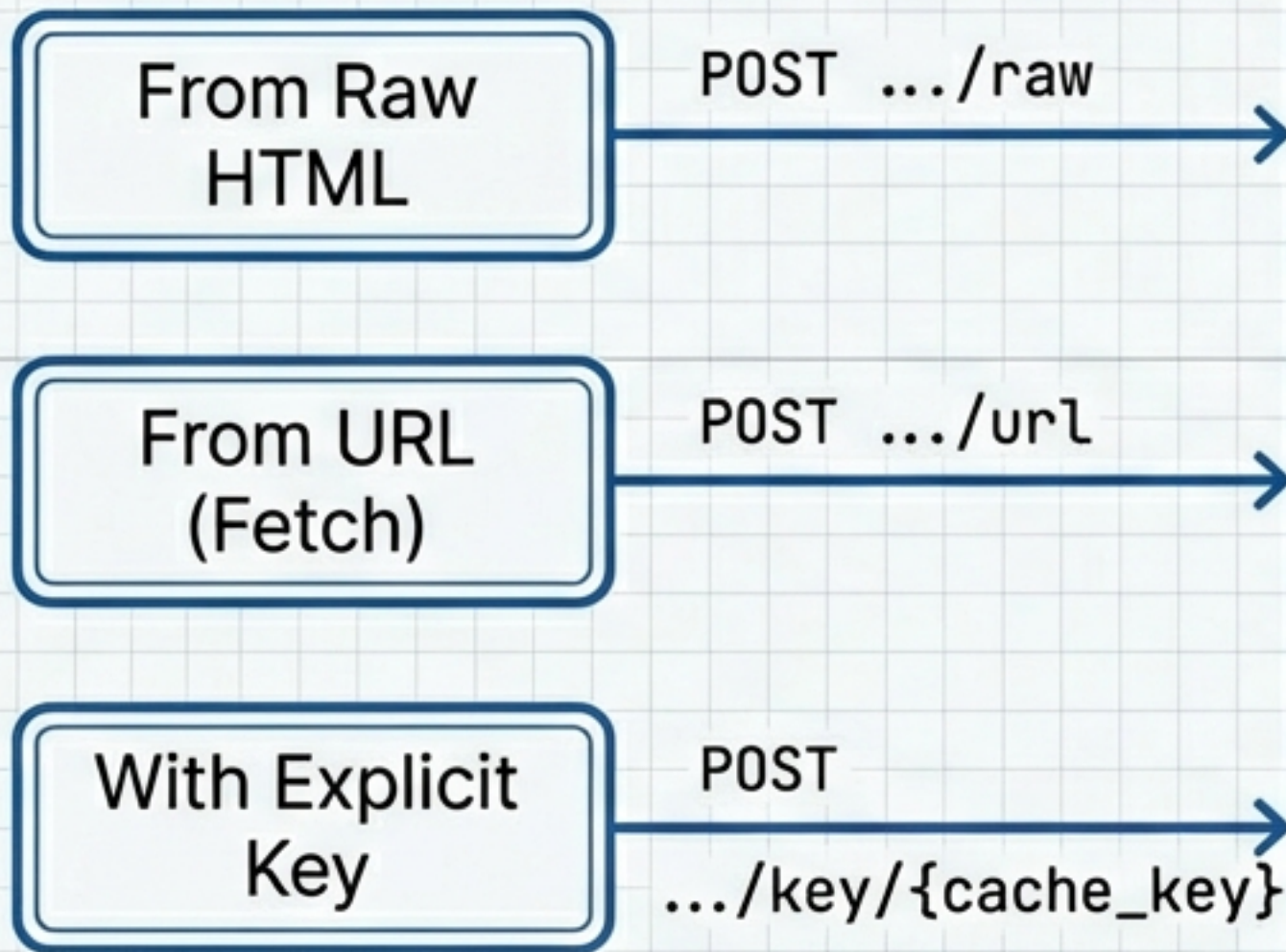
Observability Endpoints

- **GET /flows/{namespace}/{cache_id}**
- List all FLeTs associated with an ID.
- **GET ../{flet_name}/logs**
- Retrieve detailed execution logs.
- **GET ../{flet_name}/tasks**
- Breakdown of specific tasks performed.
- **GET ../{flet_name}/durations**
- Performance metrics and timing.

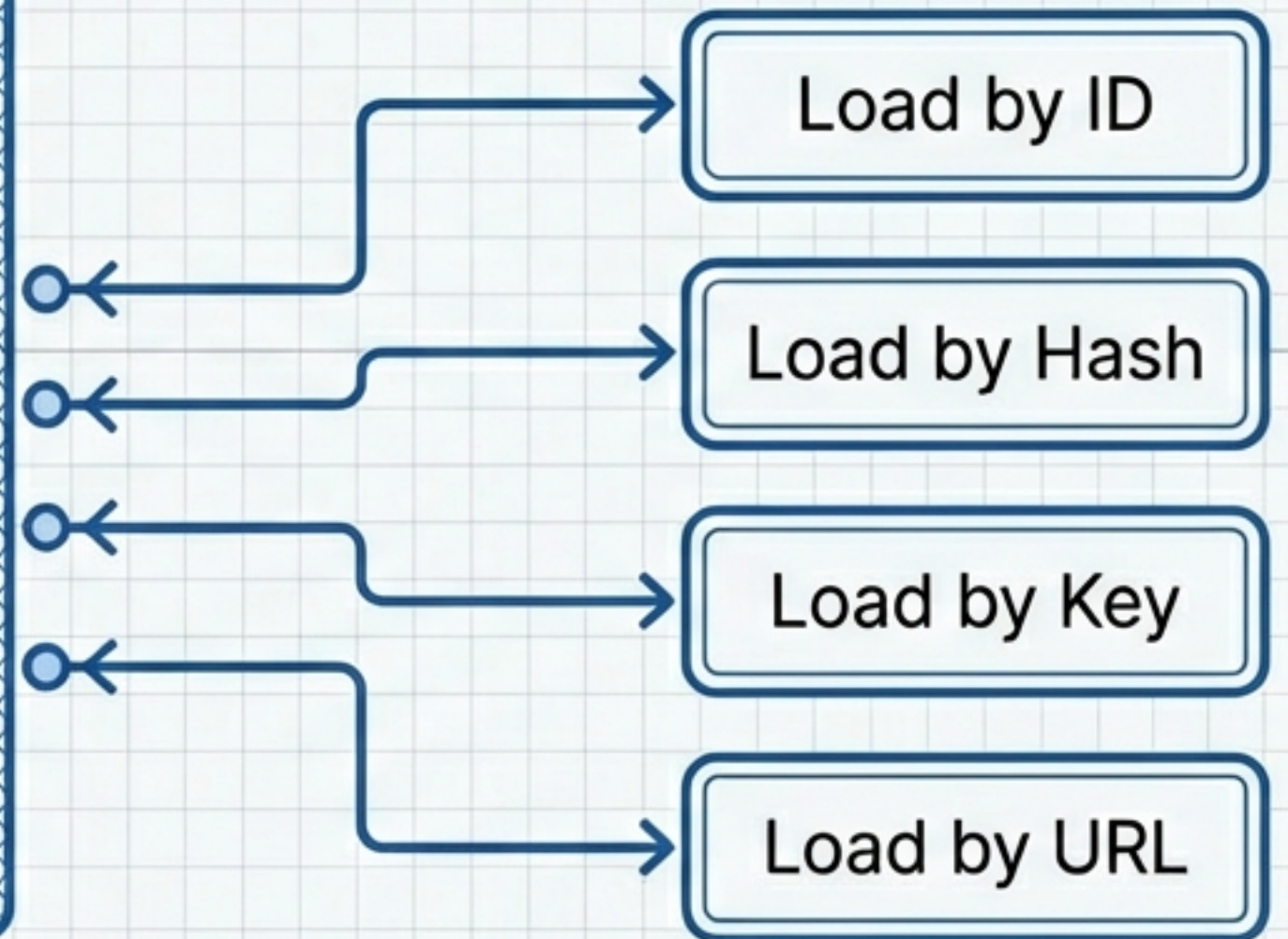
API Capability: HTML Domain Execution

Base Paths: `/flet-html-execute/` & `/flet-html-domain/`

Store Operations



Load Operations

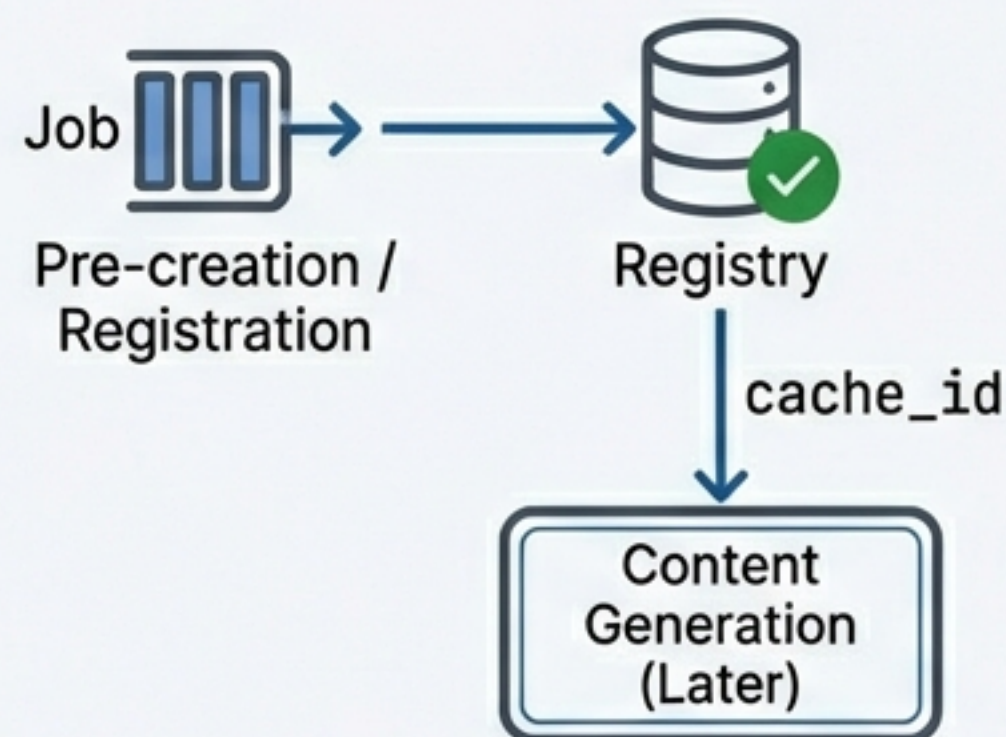


The bridge between raw cache storage and domain-specific HTML logic.

Architectural Insights & Extensibility

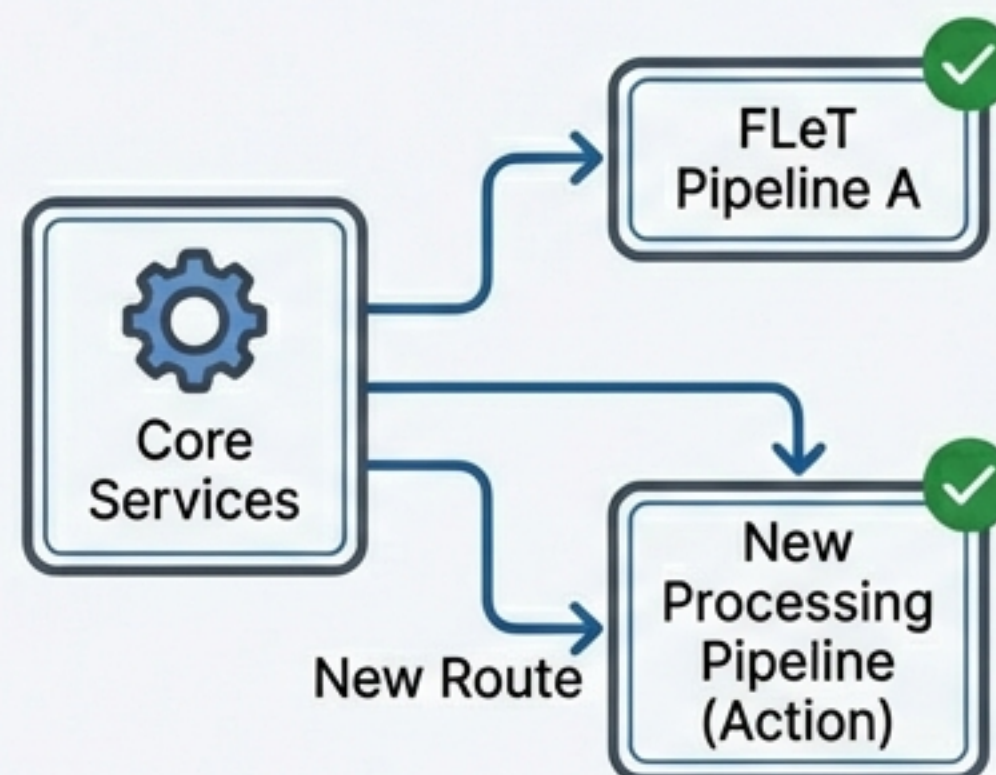
Key design principles for scalability and maintenance.

Decoupled Workflows



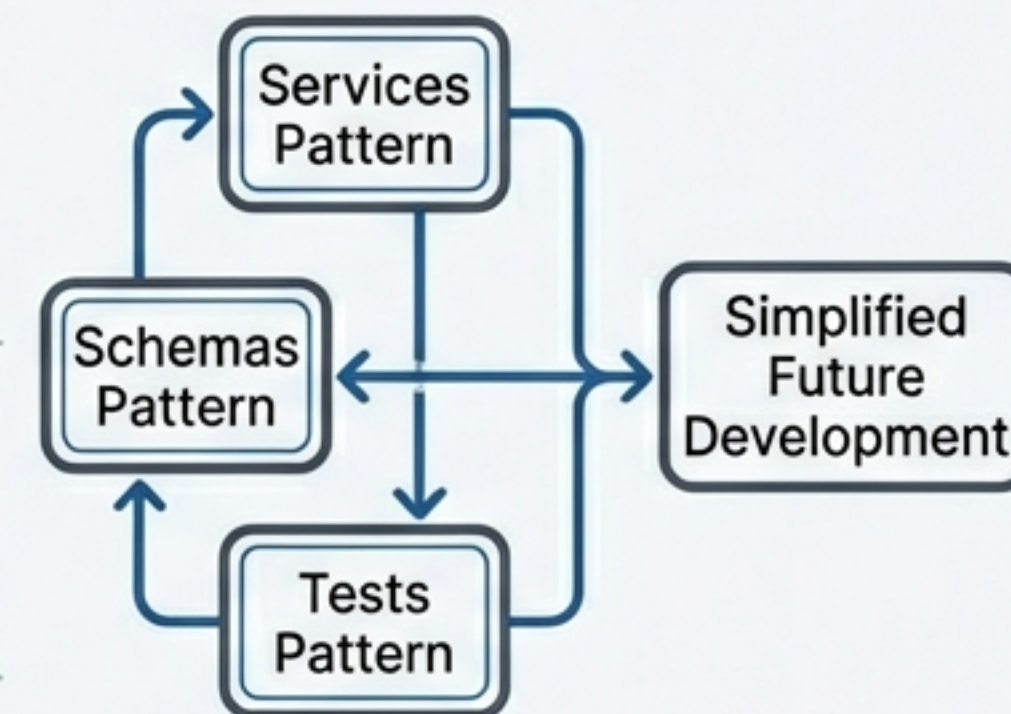
Insight: Entities can be pre-created/registered (e.g., by a job queue) before content exists. The 'cache_id' is portable and reusable.

FLeT Extensibility



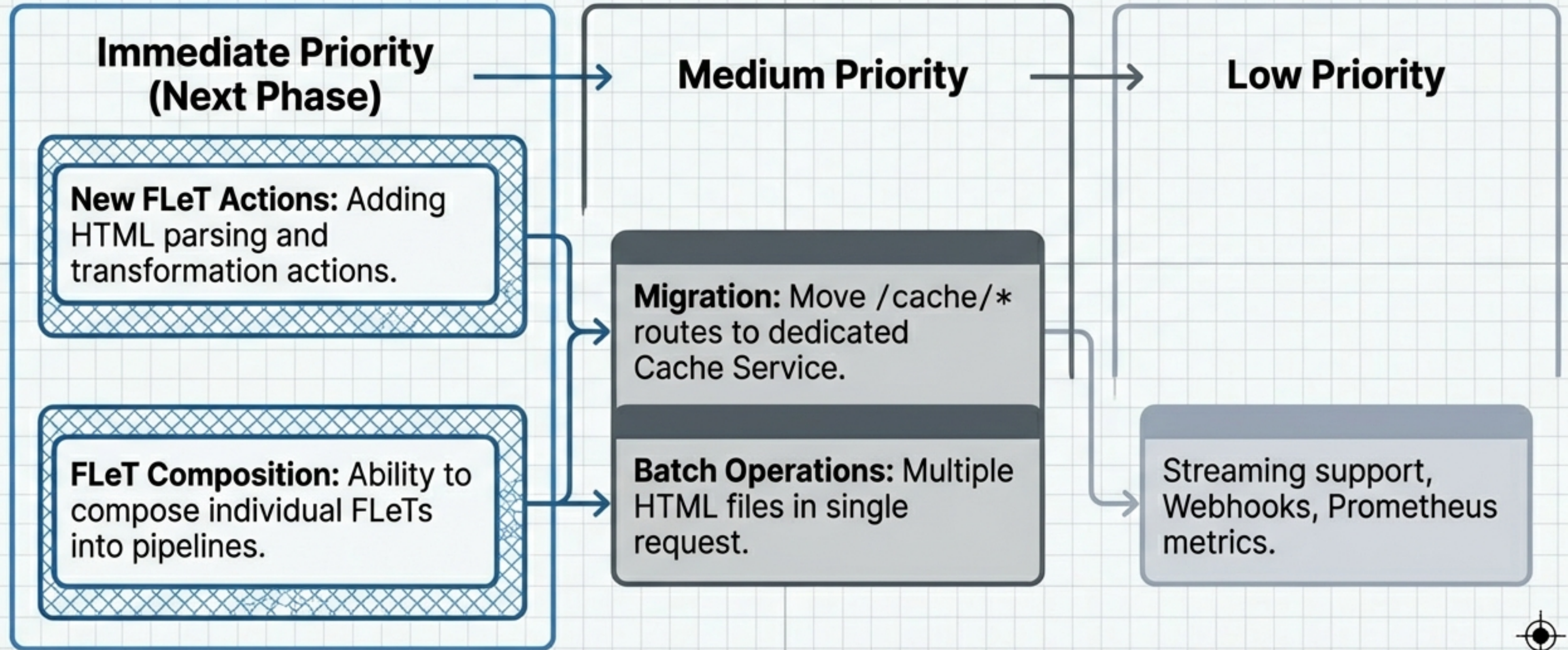
Insight: The architecture allows adding new processing pipelines (Actions) without modifying core services. Routes can expose new FLeTs with minimal wiring.

Consistent Patterns



Insight: Reusable patterns established for Services, Schemas, and Tests simplify future development.

Roadmap & Future Enhancements



Value Delivered & Resources

STAT. [3,4]

LOC. [5,1]

Impact Summary

Impact: Clients can now store, load, and inspect HTML flows with production-grade reliability.

-  **Store & Load:** From URLs, keys, or raw content.
-  **Inspect:** Full visibility into flow logs.
-  **Foundation:** A solid base for sophisticated applications.

Resources Box

Resource Links:

-  **API Base:**
<https://html-graph.dev.mgraph.ai/>
-  **Swagger Docs:** .../docs
-  **ReDoc:** .../redoc

Status:  **COMPLETE - Deployed to Production**